

Topic: C01

Security Features in Real-World Applications

1. Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:
- The security feature/service required, such as confidentiality, integrity, authentication, authorization, availability
 - The security mechanism used to provide the identified security feature.
 - The security protocol or technology used to implement the mechanism.
 - Briefly explain how the mechanism/protocol provides the identified security feature.

Application	Security Feature	Security Mechanism	Protocol/ Technology	How it provides security
Gmail	Authentication	Password + MFA	OAuth, MFA Codes/ Passkeys	Verifies the identity of the user and prevents unauthorized users from accessing the account.
Online Banking	Availability	Backup Systems	DDoS protection	Keeps banking services available to users even during failures.
Amazon (E-commerce)	Confidentiality	Encryption	HTTPS/ TLS	Protects user information while it is being sent between the user and the server.
Instagram (Social Media)	Authorization	Access Control	OAuth	Ensures that users and applications can access only the information they are permitted to use.
Google Drive (Cloud Storage)	Integrity	Hashing	SHA-256	Helps check that stored and transmitted files have not been changed or modified by an unauthorized person.
Google Classroom	Authorization	Role-Based Access Control	User Permissions	Separates teacher and student accounts so students can only view their own grades and cannot alter them.

2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Application	Security Controls	CIA	Security Mechanism	Protocol
Paytm (Digital Payment)	Transaction Verification	Integrity	OTP verification	HTTPS/ SMS Gateway(SMPP)
Zoom (Video Conferencing)	Encryption	Confidentiality	End-to-end Encryption	AES-256/ SRTP
WhatsApp (Communication)	End-to-end Encryption	Confidentiality	Cryptographic Scrambling	Signal Protocol
Netflix (Streaming Service)	Server Backup	Availability	Data Redundancy	AWS Cloud Backup
DigiLocker (Digital Document Service)	Access Control	Confidentiality	User Authorization and Access Control	OAuth/ HTTPS

Application	Prevention	Detection	Recovery
Paytm	Prevents unauthorized transactions	Detects suspicious transactions	Reverses or blocks fraudulent transactions
Zoom	Prevents unauthorized access to meetings	Detects unauthorized access	Restores secure access to the account
WhatsApp	Prevents third parties from reading messages	Detects unusual security changes	Account verification helps restore access
Netflix	Prevents service interruption	Detects server failures	Redirects users to a working server
DigiLocker	Prevents access to documents by unauthorized users	Detects suspicious login or access	Account recovery restores access